

# Optimizing a Restricted Driving Policy Across a Reduced-Speed Zone

**Abstract**—We minimize travel time, fuel cost, and expected enforcement cost for a car crossing a reduced-speed zone. A longitudinal dynamics model and an affine throttle-fuel law lead to feasible cruise speeds and computable transition costs. We first derive the model without committing to a vehicle, then give a reproducible solver for a five-stage cruise–decelerate–cruise–accelerate–cruise family. A Škoda Octavia example uses measured steady-speed fuel consumption and explicitly assumed mechanical parameters. Within that family, coasting before the zone costs less than full braking or holding the outside speed on the same route. Pontryagin conditions explain the possible arc types; the claimed minimum is restricted to the stated family.

## I. PROBLEM AND MODELING SEQUENCE

Consider a car crossing a reduced-speed zone at  $0 \leq x \leq L$ . We seek a speed profile on a fixed route that balances time, fuel, and expected enforcement cost. We first model how throttle and braking change speed, then connect throttle to measured steady-speed fuel use, and finally solve for a trajectory. The road is level, traffic-free, and without wind. Posted speed limits enter the cost as penalty thresholds, not hard constraints.

## II. VEHICLE MOTION AND FUEL USE

### A. Motion and feasible cruise

Position uses metres, time seconds, and speed m/s. Throttle  $u \in [0, 1]$  supplies propulsion, while brake command  $b \in [0, 1]$  adds deceleration  $ab$ :

$$\dot{x} = v, \quad \dot{v} = h(v)u - R(v) - ab, \quad (1)$$

$$h(v) = \frac{A}{v + v_0}, \quad A = \frac{P_w}{m}, \quad (2)$$

$$R(v) = r_0 + r_1v + Bv^2, \quad B = \frac{\rho C_D S}{2m}. \quad (3)$$

Here  $h(v)$  is full-throttle acceleration before road load,  $R(v)$  is road-load deceleration,  $P_w$  represents wheel power,  $m$  is moving mass,  $S$  is frontal area, and  $\rho$  is air density. The road-load form is empirical [2]. We specify its parameters only in Section VIII.

The small  $v_0 > 0$  keeps  $h$  finite. Propulsive power in this model is

$$P_{\text{traction}} = mh(v)uv = P_w u \frac{v}{v + v_0}. \quad (4)$$

Choose a positive speed interval  $[v_-, v_+]$  on which this power approximation is credible and  $R(v) > 0$ . It does not describe a launch from rest or individual gears.

At constant speed without braking,

$$u_{\text{cr}}(v) = \frac{R(v)}{h(v)}. \quad (5)$$

A cruise speed is feasible when  $R(v) \leq h(v)$ . An acceleration arc requires  $h(v) > R(v)$  throughout its speed range.

### B. Steady-speed fuel curve

Let  $V = 3.6v$  be speed in km/h. Let  $Q(v)$  be steady fuel use in L/h and  $Q_0$  the idling rate. Measurements of consumption per distance,  $\text{FC}_{100}(V)$  in L/100 km, become fuel rates through

$$Q(V/3.6) = \frac{V}{100} \text{FC}_{100}(V). \quad (6)$$

Fit a smooth positive  $Q$  to these rates while holding  $Q(0) = Q_0$ . A cubic interpolant is sufficient for the three moving measurements used later:

$$Q(v) = Q_0 + c_1v + c_2v^2 + c_3v^3. \quad (7)$$

The coefficients describe fuel consumption, not physical road-load terms. The solver below can use any differentiable cruise curve with  $Q(v) > Q_0$  on the selected speed interval.

### C. Fuel use during transitions

Convert cruise and idle fuel rates to L/s:

$$q_{\text{cr}}(v) = \frac{Q(v)}{3600}, \quad q_i = \frac{Q_0}{3600}. \quad (8)$$

Assume fuel use is affine in throttle. Choose its speed-dependent slope to reproduce the measured cruise curve:

$$\begin{aligned} \dot{f} &= q_i + \Gamma(v)u, \\ \Gamma(v) &= (q_{\text{cr}}(v) - q_i) \frac{h(v)}{R(v)}. \end{aligned} \quad (9)$$

Indeed, inserting (5) gives

$$q_i + \Gamma(v)u_{\text{cr}}(v) = q_{\text{cr}}(v), \quad (10)$$

Thus mechanical assumptions do not alter the fitted cruise curve. We assume  $Q(v) > Q_0$ , so  $\Gamma(v) > 0$  on the selected interval.

Cruise data alone do not determine transient fuel use. Equation (9) supplies that assumption: coasting and braking use the idle rate. It describes engine-running free coasting; fuel cut-off with engine braking would require another mode.

### III. OBJECTIVE AND OPTIMAL CONSTANT-SPEED TRAVEL

#### A. Time, fuel, and temporal enforcement

Let  $C_t > 0$  be the time cost in currency/s and  $c_f > 0$  the fuel price in currency/L. A reduced-limit zone occupies  $x \in [0, L]$ . Its regime is  $r = \text{in}$ ; elsewhere  $r = \text{out}$ . For each regime, let  $v_{\text{lim},r} < v_{\text{hi},r}$  be its two penalty thresholds. Define

$$\phi_r(v) = \begin{cases} 0, & v \leq v_{\text{lim},r}, \\ pP_{\text{low}}\lambda_r, & v_{\text{lim},r} < v \leq v_{\text{hi},r}, \\ pP_{\text{high}}\lambda_r, & v > v_{\text{hi},r}, \end{cases} \quad (11)$$

Here  $\lambda_r$  is an encounter rate in  $\text{s}^{-1}$ ,  $p \in [0, 1]$  a charge probability, and  $P_{\text{high}} > P_{\text{low}} \geq 0$  are fine amounts. The rate assumes independently chargeable encounters; the numerical rates are illustrative.

The running cost is

$$\ell_r(v, u) = C_t + c_f(q_i + \Gamma(v)u) + \phi_r(v). \quad (12)$$

For fixed  $U, W > 0$ , the general route problem minimizes

$$J = \int_0^T \ell_r(x(t))(v(t), u(t)) dt \quad (13)$$

over measurable controls satisfying (1),  $u, b \in [0, 1]$ , and  $v \in [v_-, v_+]$ , subject to

$$x(0) = -U, \quad x(T) = L + W, \quad v(0) = v(T) = v_{\text{out}}. \quad (14)$$

Terminal time  $T$  is free and there is no terminal cost. The reference speed  $v_{\text{out}}$  is selected below, then held fixed when comparing trajectories.

#### B. Cruise cost and candidate speeds

For feasible constant-speed travel,

$$g_r(v) = \frac{C_t + c_f q_{\text{cr}}(v) + \phi_r(v)}{v}, \quad (15)$$

$$v_r \in \arg \min_{\substack{v \in [v_-, v_+] \\ R(v) \leq h(v)}} g_r(v).$$

The empirical fuel data enter this calculation directly. The mechanical model only determines which cruising speeds are feasible. Within a band where  $\phi_r$  is constant, an interior stationary point satisfies

$$c_f[vq'_{\text{cr}}(v) - q_{\text{cr}}(v)] = C_t + \phi_r. \quad (16)$$

For (7), this becomes the cubic equation

$$2c_3v^3 + c_2v^2 - Q_0 = \frac{3600}{c_f}(C_t + \phi_r). \quad (17)$$

One compares admissible roots with all feasible domain endpoints, fine thresholds, and any sustainable-speed endpoint  $h(v) = R(v)$ . Thresholds are evaluated using the convention in (11); their upward jumps can create minima without satisfying (16). A sustainable-speed endpoint may be feasible for cruise but unreachable in finite time along a maximum-acceleration arc.

In the restricted construction below assume  $v_{\text{in}} < v_{\text{out}}$  and strictly positive net acceleration between them. If the selected speeds coincide, no transition is needed. If their order reverses, the assumed deceleration–acceleration sequence is inapplicable; a lower posted limit alone does not guarantee this ordering when enforcement intensities differ.

### IV. CONTROL MODES FROM THE MAXIMUM PRINCIPLE

#### A. Smooth normal extremals

On smooth intervals away from penalty thresholds and active speed constraints, use the Hamiltonian minimization convention [4]:

$$H_r = C_t + c_f q_i + \phi_r(v) + c_f \Gamma(v)u + \psi_x v + \psi_v (h(v)u - R(v) - ab). \quad (18)$$

For a normal extremal, normalize the cost multiplier to one. The costate equations are

$$\begin{aligned} \dot{\psi}_x &= 0, \\ \dot{\psi}_v &= -\phi'_r(v) - c_f \Gamma'(v)u - \psi_x - \psi_v (h'(v)u - R'(v)). \end{aligned} \quad (19)$$

The switching functions are

$$\sigma_u = c_f \Gamma(v) + \psi_v h(v), \quad \sigma_b = -a\psi_v. \quad (20)$$

Thus  $u = 0$  if  $\sigma_u > 0$  and  $u = 1$  if  $\sigma_u < 0$ ;  $b = 0$  if  $\psi_v < 0$  and  $b = 1$  if  $\psi_v > 0$ . Since  $\Gamma, h > 0$ , this gives maximum propulsion, coasting, and full braking as nonsingular modes. A switching function that stays zero over an interval may also permit a singular mode.

#### B. Singular cruise arcs

Define

$$K(v) = \frac{c_f \Gamma(v)}{h(v)} = \frac{c_f (q_{\text{cr}}(v) - q_i)}{R(v)} > 0. \quad (21)$$

On a throttle-singular interval,  $\psi_v = -K(v) < 0$ , hence  $b = 0$ . Differentiating this relation and using (19), in a band where  $\phi'_r = 0$ , yields

$$\psi_x = -[K(v)R(v)]' = -c_f q'_{\text{cr}}(v). \quad (22)$$

Since  $\psi_x$  is constant,  $q''_{\text{cr}}(v)\dot{v} = 0$ . If the supplied cruise curve is strictly convex, as in the example, a smooth throttle-singular arc has constant speed and throttle  $u = R/h$ . For the autonomous free-terminal-time problem,  $H = 0$  on a normal extremal [4]. Substituting (22) in (18) then recovers (16).

The singular conditions are necessary, not a global optimality test. A smooth brake-singular interval would give  $\psi_v = 0$ , then  $u = 0$  and  $\psi_x = 0$ . Its Hamiltonian would be  $C_t + c_f q_i + \phi_r > 0$ , contradicting  $H = 0$  for the normal free-time problem.

### C. From modes to a computable family

At a fine threshold, the penalty jumps and smooth stationarity does not apply. We evaluate threshold cruises directly through  $g_r$ . The maximum principle identifies useful modes but does not fix their order or prove a global optimum. We now prescribe a five-stage family and optimize its switching locations. Mixed coast-brake deceleration and extra switches lie outside that family.

### V. FIVE-STAGE FAMILY AND TRANSITION QUADRATURES

Fix feasible reference speeds  $v_{\text{in}} < v_{\text{out}}$  from (15). The five stages are: outside cruise at  $v_{\text{out}}$ ; deceleration to  $v_{\text{in}}$ ; inside cruise at  $v_{\text{in}}$ ; maximum acceleration back to  $v_{\text{out}}$ ; and outside cruise at  $v_{\text{out}}$ . Deceleration is either entirely coasting or entirely full braking. Zero-length cruise stages are allowed. We require

$$\begin{aligned} R_d(v) &= R(v) + ab_d > 0, \quad b_d \in \{0, 1\}, \\ F(v) &= h(v) - R(v) > 0 \end{aligned} \quad (23)$$

on  $[v_{\text{in}}, v_{\text{out}}]$ . The total distances are

$$D_d = \int_{v_{\text{in}}}^{v_{\text{out}}} \frac{v}{R_d(v)} dv, \quad D_a = \int_{v_{\text{in}}}^{v_{\text{out}}} \frac{v}{F(v)} dv. \quad (24)$$

They follow from  $dx/dv = v/\dot{v}$ .

#### A. Boundary-speed parametrization

Require the deceleration arc to touch or cross  $x = 0$  and the acceleration arc to touch or cross  $x = L$ . Define

$$\begin{aligned} v_b &= v(x = 0), \quad v_c = v(x = L), \\ (v_b, v_c) &\in [v_{\text{in}}, v_{\text{out}}]^2. \end{aligned} \quad (25)$$

These are spatial evaluations, not the initial and terminal speeds at times 0 and  $T$ . The split distances are

$$\begin{aligned} D_d^{\text{out}}(v_b) &= \int_{v_b}^{v_{\text{out}}} \frac{v}{R_d(v)} dv, \quad D_d^{\text{in}}(v_b) = \int_{v_{\text{in}}}^{v_b} \frac{v}{R_d(v)} dv, \\ D_a^{\text{in}}(v_c) &= \int_{v_{\text{in}}}^{v_c} \frac{v}{F(v)} dv, \quad D_a^{\text{out}}(v_c) = \int_{v_c}^{v_{\text{out}}} \frac{v}{F(v)} dv. \end{aligned} \quad (26)$$

Their sums are  $D_d$  and  $D_a$ , independent of boundary speeds for fixed reference speeds and deceleration mode. The switching positions are

$$\begin{aligned} x_{12} &= -D_d^{\text{out}}, & x_{23} &= D_d^{\text{in}}, \\ x_{34} &= L - D_a^{\text{in}}, & x_{45} &= L + D_a^{\text{out}}. \end{aligned} \quad (27)$$

The inside-cruise length is

$$L_{\text{cr}} = L - D_d^{\text{in}}(v_b) - D_a^{\text{in}}(v_c) \geq 0. \quad (28)$$

This family includes the four endpoint combinations, but also allows intermediate boundary speeds. It does not include arbitrary translations of a transition entirely away from its associated boundary.

### B. Transition costs

Define deceleration and acceleration cost integrands by

$$\begin{aligned} L_{d,r}(v) &= C_t + c_f q_i + \phi_r(v), \\ L_{a,r}(v) &= C_t + c_f (q_i + \Gamma(v)) + \phi_r(v). \end{aligned} \quad (29)$$

Then

$$\begin{aligned} J_d^{\text{out}}(v_b) &= \int_{v_b}^{v_{\text{out}}} \frac{L_{d,\text{out}}(v)}{R_d(v)} dv, \\ J_d^{\text{in}}(v_b) &= \int_{v_{\text{in}}}^{v_b} \frac{L_{d,\text{in}}(v)}{R_d(v)} dv, \\ J_a^{\text{in}}(v_c) &= \int_{v_{\text{in}}}^{v_c} \frac{L_{a,\text{in}}(v)}{F(v)} dv, \\ J_a^{\text{out}}(v_c) &= \int_{v_c}^{v_{\text{out}}} \frac{L_{a,\text{out}}(v)}{F(v)} dv. \end{aligned} \quad (30)$$

These are one-dimensional quadratures. Split numerical integration at every fine threshold to avoid integrating blindly across a jump. The additional speed dependence of  $\Gamma$  changes the acceleration integrand but does not increase the optimization dimension.

### VI. A COMMON ROUTE AND A FINITE-DIMENSIONAL OPTIMIZATION

Let  $G_{\text{out}} = g_{\text{out}}(v_{\text{out}})$  and  $G_{\text{in}} = g_{\text{in}}(v_{\text{in}})$ . Choose the same  $U, W$  for every comparison, with

$$U \geq \max_{b_d \in \{0,1\}} D_d, \quad W \geq D_a. \quad (31)$$

The outside cruise distances are  $U - D_d^{\text{out}}$  before deceleration and  $W - D_a^{\text{out}}$  after acceleration. Therefore

$$\begin{aligned} J_d(v_b, v_c) &= G_{\text{out}}(U - D_d^{\text{out}}) + G_{\text{out}}(W - D_a^{\text{out}}) \\ &\quad + G_{\text{in}}(L - D_d^{\text{in}} - D_a^{\text{in}}) \\ &\quad + J_d^{\text{out}} + J_d^{\text{in}} + J_a^{\text{in}} + J_a^{\text{out}}. \end{aligned} \quad (32)$$

All split quantities on the right are evaluated at their corresponding boundary speed as defined in (26) and (30). Minimize this expression over (28) separately for each deceleration mode, then compare the two minima. Evaluating each mode on its own window  $[-D_d, L + D_a]$  is invalid for the cross-mode comparison, since full braking traverses a shorter route. If such mode-specific costs have already been computed, add

$$G_{\text{out}}[(U - D_d) + (W - D_a)] \quad (33)$$

to obtain costs on the common route.

#### A. Interior candidates and discontinuity candidates

For fixed mode, (32) separates into a constant, a function of  $v_b$ , and a function of  $v_c$ . Define

$$\begin{aligned} \Delta\phi(v) &= \phi_{\text{in}}(v) - \phi_{\text{out}}(v), \\ N(v) &= (G_{\text{out}} - G_{\text{in}})v + \Delta\phi(v). \end{aligned} \quad (34)$$

Away from thresholds,

$$\frac{\partial J_d}{\partial v_b} = \frac{N(v_b)}{R_d(v_b)}, \quad \frac{\partial J_d}{\partial v_c} = \frac{N(v_c)}{F(v_c)}. \quad (35)$$

All fuel and time terms on a given transition cancel in this derivative because moving the regime boundary changes only its fine rate and the displaced cruise distances. Thus the boundary-speed equations retain the same simple form even with speed-dependent throttle fuel use.

In each common fine band, a stationary candidate is

$$v^* = -\frac{\Delta\phi}{G_{\text{out}} - G_{\text{in}}}, \quad (36)$$

provided the denominator is nonzero and the root lies in the band and in  $[v_{\text{in}}, v_{\text{out}}]$ . Add the interval endpoints and every fine threshold in that interval. The transition-cost integrals themselves are continuous as functions of their limits, but have derivative changes at the thresholds.

If  $G_{\text{out}} = G_{\text{in}}$ , derivatives have the sign of  $\Delta\phi$  within each band; if  $\Delta\phi = 0$  as well, that band's contribution is flat. Flat candidate intervals must be retained, rather than discarded by division by zero. If independent global minimizers of the two separated terms are feasible, their combination is globally optimal within the fixed-mode family. Otherwise evaluate all feasible candidate combinations and the active constraint below. First-order roots alone do not guarantee minima.

### B. Active inside-length constraint

On the active boundary,

$$D_d^{\text{in}}(v_b) + D_a^{\text{in}}(v_c) = L. \quad (37)$$

For every  $v_b$  for which  $L - D_d^{\text{in}}(v_b) \in [0, D_a]$ , determine the unique  $v_c(v_b)$  by inversion of the strictly increasing function  $D_a^{\text{in}}$ . The problem becomes a one-dimensional minimization of  $J_d(v_b, v_c(v_b))$ . Split this search at both entry and exit fine-threshold events and include its endpoints. On a smooth portion,

$$\frac{d}{dv_b} J_d(v_b, v_c(v_b)) = \frac{\Delta\phi(v_b) - (v_b/v_c)\Delta\phi(v_c)}{R_d(v_b)}. \quad (38)$$

This supplies additional stationary candidates or identifies a flat portion. Comparing all these candidates gives the global minimum of the restricted two-variable problem, including its boundary. The feasible set is compact and the objective is continuous under the stated positive-denominator assumptions, so a restricted minimum exists.

An active constraint means only that the prescribed inside cruise has zero length. It does not prove that an unrestricted short-zone optimum reaches  $v_{\text{in}}$ . A useful extension is to optimize a turning speed  $w \in [v_{\text{in}}, v_{\text{out}}]$ , replace the lower integration limits by  $w$ , and use  $g_{\text{in}}(w)$  for any inside plateau. Another is to add a coast-brake switching speed. Both remain finite-dimensional restrictions, without establishing unrestricted global optimality. Constant-speed travel at  $v_{\text{out}}$  should also be compared as a separate baseline.

### C. Solver procedure

The formulas above give a direct numerical implementation for any supplied  $h$ ,  $R$ ,  $Q$ , and penalty rates:

- 1) Restrict speeds to  $[v_-, v_+]$  and retain feasible cruise speeds with  $R \leq h$ . For each regime, minimize  $g_r$ : solve (16) within each constant-penalty band and compare its roots with band endpoints, thresholds, and feasible-domain endpoints. Select  $v_{\text{out}}$  and  $v_{\text{in}}$ .
- 2) If  $v_{\text{in}} < v_{\text{out}}$ , check  $R_d > 0$  and  $F > 0$  on their interval. For each  $b_d \in \{0, 1\}$ , compute (26) and (30) with one-dimensional quadrature. Break each integral at any fine threshold inside it.
- 3) Search  $[v_{\text{in}}, v_{\text{out}}]^2$  subject to (28). Evaluate endpoint and threshold pairs, smooth stationary candidates from (35), and the active boundary. On that boundary, invert  $D_a^{\text{in}}$  for  $v_c$  at each  $v_b$  and perform a one-dimensional search using (38) between its threshold events. Evaluate (32) at every feasible candidate; retain the smallest value for each mode.
- 4) Choose common  $U, W$  large enough for both modes. Compare their minima and a constant- $v_{\text{out}}$  baseline using the same route,  $[-U, L + W]$ . Recompute time, fuel, and fines separately as a numerical check.

If the selected cruise speeds coincide or reverse order, use a different stage order. If a short zone makes the prescribed inside speed unattractive, add the turning speed  $w$  as a search variable.

## VII. AN EXACT IDENTITY FOR CHECKING COSTS

Let  $\Xi'(v) = K(v)$  with  $K$  from (21). Eliminating throttle from (1) and (9) gives

$$\begin{aligned} J = & \Xi(v(T)) - \Xi(v(0)) \\ & + \int_{x(0)}^{x(T)} g_{r(x)}(v(x)) dx \\ & + \int_0^T K(v(t)) ab(t) dt. \end{aligned} \quad (39)$$

Indeed  $c_f \Gamma u = K\dot{v} + KR + Kab$ , while  $KR = c_f(q_{\text{cr}} - q_i)$ . With equal initial and final speeds, the endpoint term cancels. This identity is an algebraic property of the assumed affine fuel model, not a general law of vehicle energetics. It provides an independent check of time-domain or quadrature-based cost calculations and displays the explicit braking penalty. It does not imply that braking can never be optimal: braking can change the speed profile, fines, travel time, and the available transition distance.

## VIII. NUMERICAL EXAMPLE

### A. Vehicle data and model parameters

The example uses owner-reported steady-speed consumption for a facelifted Škoda Octavia III Combi 2.0 TDI. The measured car's transmission, model year, and loaded mass are unspecified. The manufacturer reports the 110 kW rating and  $C_D = 0.302$  for 2017 front-wheel-drive variants [1]; these figures do not determine steady-speed fuel use.

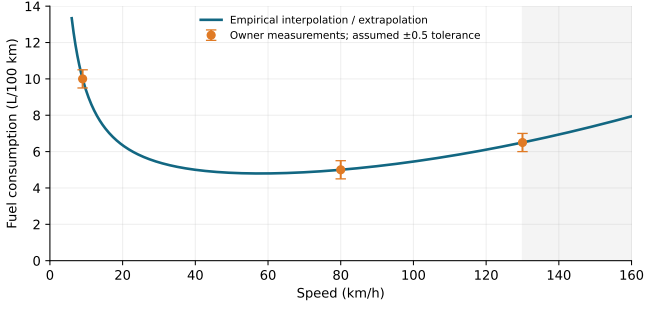


Fig. 1. Measured steady-speed consumption and cubic fit. The region above 130 km/h is extrapolation. Error bars show an illustrative  $\pm 0.5$  L/100 km tolerance, not measured uncertainty.

TABLE I  
MECHANICAL INPUTS FOR THE NUMERICAL EXAMPLE.

| Parameter          | Value                                         | Basis         |
|--------------------|-----------------------------------------------|---------------|
| $P_e; C_D$         | 110 kW; 0.302                                 | Published [1] |
| $m; \eta_d$        | 1500 kg; 0.90                                 | Assumed       |
| $P_w = \eta_d P_e$ | 99 kW                                         | Derived       |
| $S; \rho$          | 2.20 m <sup>2</sup> ; 1.225 kg/m <sup>3</sup> | Assumed       |
| $C_{rr}; g$        | 0.012; 9.81 m/s <sup>2</sup>                  | Assumed       |
| $v_0; a$           | 1 m/s; 7 m/s <sup>2</sup>                     | Assumed       |

The owner-reported data are

$$\begin{array}{c|ccc} V \text{ (km/h)} & 9 & 80 & 130 \\ \hline \text{FC}_{100} \text{ (L/100 km)} & 10.0 & 5.0 & 6.5 \end{array} \quad (40)$$

with idle rate  $Q_0 = 0.6$  L/h. Equations (6)–(7) convert the moving points to 0.9, 4.0, and 8.45 L/h and give

$$\begin{aligned} c_1 &= 1.20713517724 \times 10^{-1}, \\ c_2 &= -5.05754823921 \times 10^{-4}, \\ c_3 &= 8.81390936848 \times 10^{-5}. \end{aligned} \quad (41)$$

Here  $v$  is in m/s and  $Q$  is in L/h. The cubic passes through all four rates. Its negative  $c_2$  is an interpolation coefficient, not a negative resistance. On the chosen interval,  $Q(v) > Q_0$  and  $Q''(v) > 0$ . Above 130 km/h it extrapolates beyond the measured moving speeds.

For mechanics, set  $r_0 = C_{rr}g$  and  $r_1 = 0$ . Table I gives every input. Only  $P_e$  and  $C_D$  come from the cited manufacturer specification; the other vehicle parameters are scenario assumptions. These values give  $A = 66.0$  m<sup>2</sup>/s<sup>3</sup>,  $B = 2.712966667 \times 10^{-4}$  m<sup>-1</sup>, and  $r_0 = 0.11772$  m/s<sup>2</sup>. We solve on  $60 \leq V \leq 160$  km/h. At 130 km/h the assumed road load needs about 25.54 kW at the wheels. Combined with 8.45 L/h and a diesel heating value of 35.8 MJ/L [3], this implies about 30.4% fuel-to-wheel efficiency as a consistency check. The transient fuel law predicts about 30.21 L/h at full throttle there.

The coast model predicts 6.25 s for 130-to-120 km/h. An earlier rough observation was 3–6 s, but gear state, gradient, and wind were unspecified. Transition results below therefore depend on the assumed mechanics.

## B. Economic inputs and reference speeds

Use the mechanical inputs in Table I, the fuel interpolation (7), and

$$\begin{aligned} L &= 1000 \text{ m}, & U &= W = 2000 \text{ m}, \\ C_t &= 300/3600 \text{ CZK/s}, & c_f &= 32 \text{ CZK/L}, \\ p &= 1, & \lambda_{\text{out}} &= \lambda_{\text{in}} = 0.026/3600 \text{ s}^{-1}, \\ P_{\text{low}} &= 2500 \text{ CZK}, & P_{\text{high}} &= 15000 \text{ CZK}. \end{aligned} \quad (42)$$

The outside thresholds are 130 and 160 km/h, and the inside thresholds are 80 and 110 km/h. The two nonzero fine rates are consequently 65/3600 and 390/3600 CZK/s. All variants cover the same 5 km route from  $x = -2000$  to  $x = 3000$  m.

Minimizing (15) on the assumed 60–160 km/h driving interval gives

$$v_{\text{out}} = 130 \text{ km/h}, \quad v_{\text{in}} = 110 \text{ km/h}, \quad (43)$$

with

$$\begin{aligned} 1000G_{\text{out}} &= 4.38769 \text{ CZK/km}, \\ 1000G_{\text{in}} &= 5.15984 \text{ CZK/km}. \end{aligned} \quad (44)$$

At 80 km/h the inside cruise cost is 5.35000 CZK/km. The model therefore chooses 110 km/h in the nominal 80 km/h zone. At exactly 110 km/h the lower fine applies; immediately above it the higher fine applies. This threshold convention is consequential. Both selected cruise speeds are threshold minima rather than roots of the smooth cruise-stationarity equation.

For  $110 < V < 130$  km/h,  $\Delta\phi = 390/3600$  CZK/s and  $N(v) > 0$ . Hence both partial derivatives in (35) are positive throughout the open interval. For either deceleration mode, the restricted minimum is

$$v_b^* = v_c^* = 110 \text{ km/h}. \quad (45)$$

Deceleration finishes at the zone entrance and acceleration starts at its exit. The inside-length constraint is inactive. This proves the boundary-speed minimum for this example, without asserting that the restricted sequence is the global solution of (13).

## C. Costs, distances, and interpretation

Table II gives the common-route comparison. The unchanged-speed baseline remains at 130 km/h throughout, including the zone, and is outside the five-stage family with a prescribed lower cruise speed.

Relative to full braking, the coasting profile takes about 1.01 s longer and consumes about 0.02500 L less fuel. The time penalty is about 0.08382 CZK, the fuel saving about 0.79993 CZK, and the net saving about 0.71611 CZK. Relative to unchanged-speed travel, it takes about 6.37 s longer and lowers the modeled total cost by about 2.18621 CZK.

TABLE II  
COMPUTED RESULTS ON THE SAME 5 KM ROUTE.

| Quantity             | Coast           | Brake    | 130 km/h |
|----------------------|-----------------|----------|----------|
| Decel. distance (m)  | 441.71          | 24.95    | —        |
| Accel. distance (m)  | 124.15          | 124.15   | —        |
| Travel time (s)      | 144.83          | 143.83   | 138.46   |
| Fuel (L)             | 0.31537         | 0.34037  | 0.32500  |
| Expected fines (CZK) | 0.59091         | 0.59091  | 3.00000  |
| Total cost (CZK)     | <b>22.75225</b> | 23.46836 | 24.93846 |

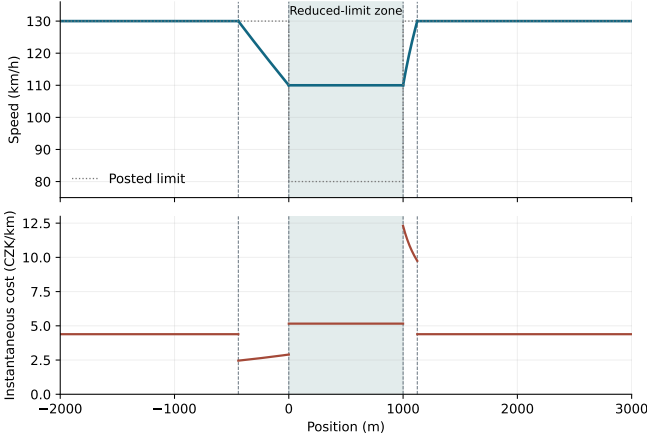


Fig. 2. Optimal coasting profile within the five-stage family. The shaded zone has an 80 km/h posted limit; the model chooses 110 km/h. Dashed lines mark control changes.

The coasting switching positions are approximately

$$(x_{12}, x_{23}, x_{34}, x_{45}) = (-441.71, 0, 1000, 1124.15) \text{ m.} \quad (46)$$

The profile is shown in Figure 2. Its instantaneous cost per distance is  $1000\ell_r(v, u)/v$  in CZK/km. This differs from the cruise density  $1000g_r(v)$  on acceleration and deceleration arcs. Jumps at control changes are expected in the affine model; isolated endpoint values do not affect the integrals.

#### IX. LIMITATIONS AND MEASUREMENTS THAT WOULD IMPROVE THE MODEL

The fuel interpolation, the assumed road load, and the assumed transient fuel law play different roles. Changing the road load or wheel-power parameter while retaining (9) preserves the measured cruise curve and its unconstrained economic minimizers, provided those speeds remain feasible. It changes transition distances, transition fuel use, and the total route cost. Agreement at the three cruise points cannot select among such mechanical models.

The most useful additional data would be a known loaded mass, repeated level-road coast-down times with gear and clutch state recorded, and timed acceleration with simultaneous fuel measurements over the actual transition-speed range. Fuel cut-off and engine braking should be introduced together as a separate control mode if they are important in the measured vehicle. An energy-efficiency

map or full transmission model is not needed merely to optimize the present restricted family.

The example’s numerical precision serves reproducibility, not a claim of equally precise prediction. Uncertainty in fuel measurements, road load, enforcement intensity, and the value of time can change the selected policy. The sharp fine thresholds can also favor exact-threshold speeds; a smoothed or uncertain threshold would be a different modeling choice.

#### X. CONCLUSION

Direct interpolation of measured fuel consumption avoids an unsupported conversion from fuel use to engine power or vehicle mass. A speed-dependent coefficient in an otherwise affine throttle law preserves the mathematical structure needed for simple cruise optimization, local PMP calculations, and transition quadratures. Smooth singular-cruise conditions can be derived explicitly, while fine-threshold cruises and the choice of a five-stage sequence are treated separately. Optimizing two boundary speeds gives an exact minimum within the prescribed family, and a fixed spatial window permits meaningful comparison between coasting, braking, and baseline travel. Under the illustrative inputs, coasting to 110 km/h before the zone and accelerating after it has the lowest cost among the compared alternatives; its transition distances remain conditional on unmeasured mechanical and transient-fuel assumptions.

#### APPENDIX REPRODUCIBILITY

The accompanying `reproduce.py` requires Python 3, NumPy, SciPy, and Matplotlib. It solves the fuel interpolation, enumerates cruise candidates, evaluates the transition quadratures, checks the cost decomposition and identity, and generates the figures and `results.json`. From this source directory, compile with `latexmk -pdf article.tex`. Pre-generated PDF figures are supplied, so Python is not needed to compile the article.

#### REFERENCES

- [1] ŠKODA AUTO, *87th Geneva International Motor Show: Press Kit*, 7 March 2017, PDF pp. 59–61, “ŠKODA OCTAVIA COMBI: Diesel engines.” <https://cdn.skoda-storyboard.com/2017/03/170307-Geneva-2017-Press-Kit.pdf>.
- [2] U.S. Environmental Protection Agency, *Population and Activity of Onroad Vehicles in MOVES5*, Section 15, “Vehicle Mass and Road Load Coefficients.” <https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockey=P101CUN7.TXT>.
- [3] U.S. Department of Energy, Alternative Fuels Data Center, *Fuel Properties Comparison*, low-sulfur diesel lower heating value (approximately 35.8 MJ/L). <https://afdc.energy.gov/fuels/properties>.
- [4] D. Liberzon, *Calculus of Variations and Optimal Control Theory: A Concise Introduction*. Princeton University Press, 2012. Sections 4.1.1 and 4.4.3; author’s online version: <https://liberzon.csl.illinois.edu/teaching/cvoc/node63.html> and <https://liberzon.csl.illinois.edu/teaching/cvoc/node87.html>.